

ASSIGNMENT NO.9: Data Retrieval, Functions, and Joins in MySQL

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: OnlineEnrollment and OfflineEnrollment.

OnlineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

OfflineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	
Column Name	Data Type	Constraint
City	VARCHAR(30)	
Marks	INT	

Insert at least 6 records into each table. Some records should be common in both tables.

Answer:-

```
CREATE TABLE OnlineEnrollment (
```

```
    StudentID INT PRIMARY KEY,
```

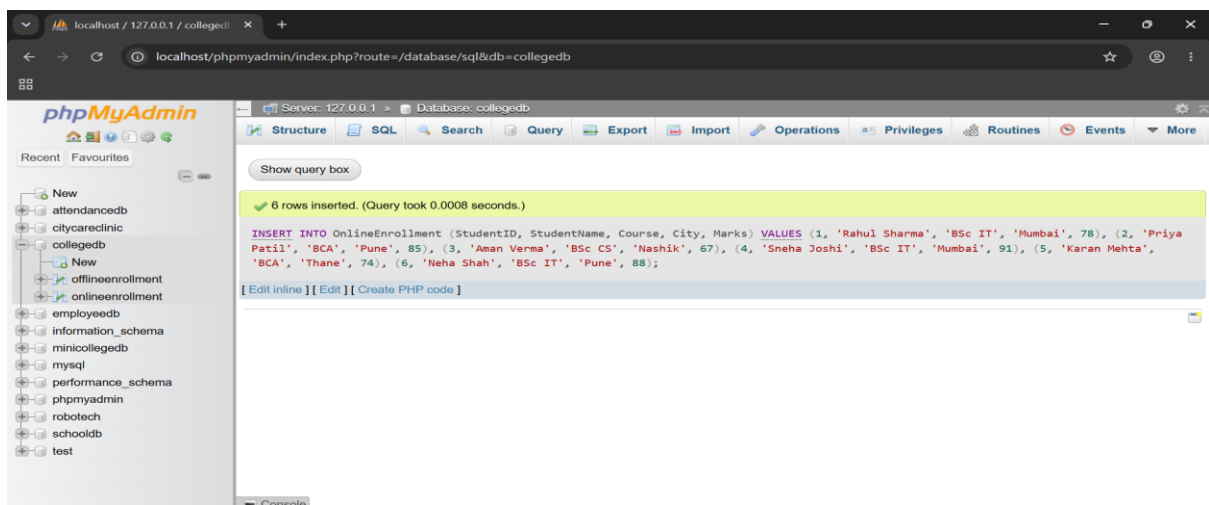
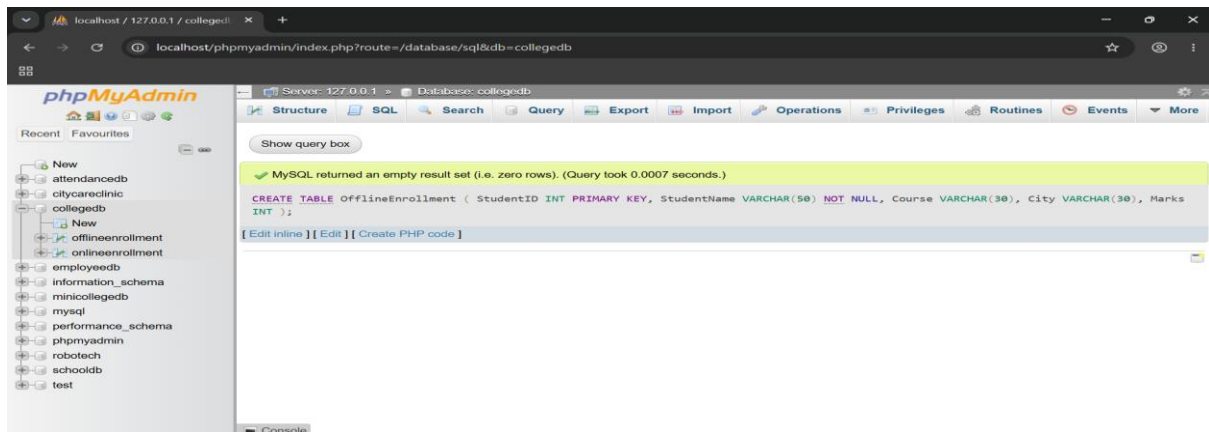
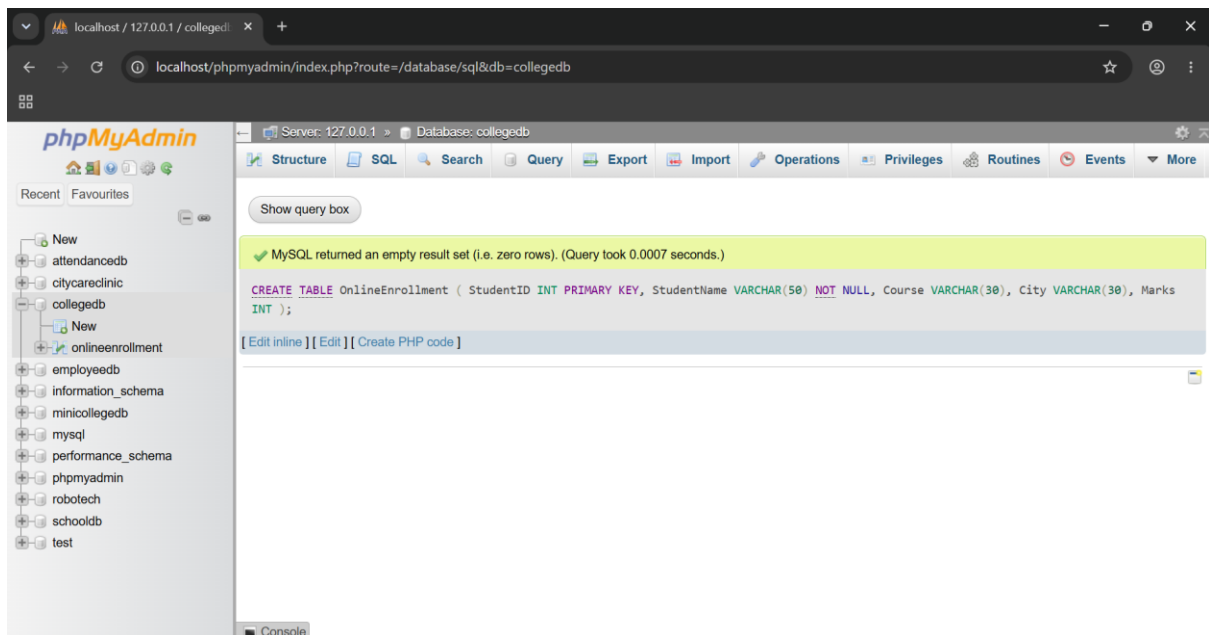
```
    StudentName VARCHAR(50) NOT NULL,
```

```
    Course VARCHAR(30),
```

City VARCHAR(30),

Marks INT

);



localhost / 127.0.0.1 / colleged: x +

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

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Server: 127.0.0.1 > Database: collegedb

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Show query box

6 rows inserted. (Query took 0.0050 seconds.)

```
INSERT INTO OfflineEnrollment (StudentID, StudentName, Course, City, Marks) VALUES (4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91), (5, 'Karan Mehta', 'BCA', 'Thane', 74), (7, 'Rohan Desai', 'BSc CS', 'Mumbai', 69), (8, 'Meera Kulkarni', 'BCA', 'Nashik', 82), (9, 'Arjun Nair', 'BSc IT', 'Thane', 59), (10, 'Divya More', 'BCA', 'Mumbai', 93);
```

[Edit inline] [Edit] [Create PHP code]

Console

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localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

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Server: 127.0.0.1 > Database: collegedb > Table: OnlineEnrollment

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

SELECT * FROM OnlineEnrollment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentID	StudentName	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	1	Rahul Sharma	BSc IT	Mumbai	78
☐ Edit ☐ Copy ☐ Delete	2	Priya Patil	BCA	Pune	85
☐ Edit ☐ Copy ☐ Delete	3	Aman Verma	BSc CS	Nashik	67
☐ Edit ☐ Copy ☐ Delete	4	Sneha Joshi	BSc IT	Mumbai	91
☐ Edit ☐ Copy ☐ Delete	5	Karan Mehta	BCA	Thane	74
☐ Edit ☐ Copy ☐ Delete	6	Neha Shah	BSc IT	Pune	88

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Console

localhost / 127.0.0.1 / colleged: x +

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=OnlineEnrollment

phpMyAdmin

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- test

Server: 127.0.0.1 > Database: collegedb > Table: OfflineEnrollment

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

SELECT * FROM OfflineEnrollment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentID	StudentName	Course	City	Marks
☐ Edit ☐ Copy ☐ Delete	4	Sneha Joshi	BSc IT	Mumbai	91
☐ Edit ☐ Copy ☐ Delete	5	Karan Mehta	BCA	Thane	74
☐ Edit ☐ Copy ☐ Delete	7	Rohan Desai	BSc CS	Mumbai	69
☐ Edit ☐ Copy ☐ Delete	8	Meera Kulkarni	BCA	Nashik	82
☐ Edit ☐ Copy ☐ Delete	9	Arjun Nair	BSc IT	Thane	59
☐ Edit ☐ Copy ☐ Delete	10	Divya More	BCA	Mumbai	93

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Console

2) Write a MySQL query using UNION to display all unique students from both OnlineEnrollment and OfflineEnrollment.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

Answer:-

MySQL query

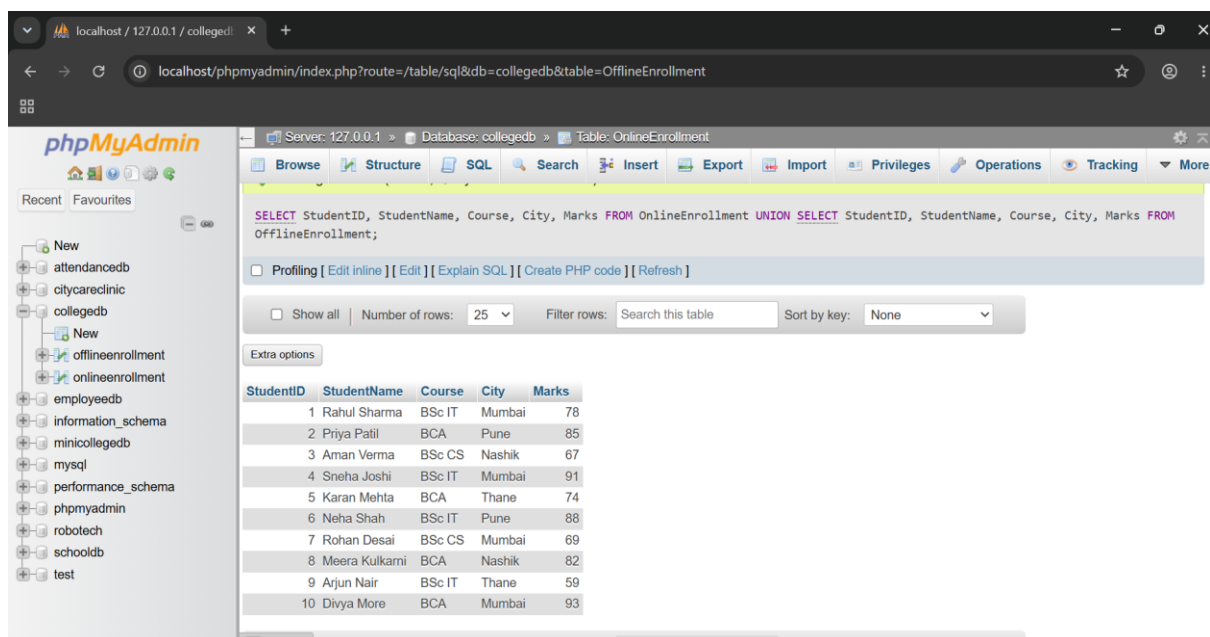
SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

UNION

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;



The screenshot shows the phpMyAdmin interface with the following query executed:

```
SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment UNION SELECT StudentID, StudentName, Course, City, Marks FROM OfflineEnrollment;
```

The result table displays the following data:

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
6	Neha Shah	BSc IT	Pune	88
7	Rohan Desai	BSc CS	Mumbai	69
8	Meera Kulkarni	BCA	Nashik	82
9	Arjun Nair	BSc IT	Thane	59
10	Divya More	BCA	Mumbai	93

Q3) Write a MySQL query using UNION ALL to combine records from both tables while keeping duplicate records.

Also write one line explaining the difference between UNION and UNION ALL.

Answer:-

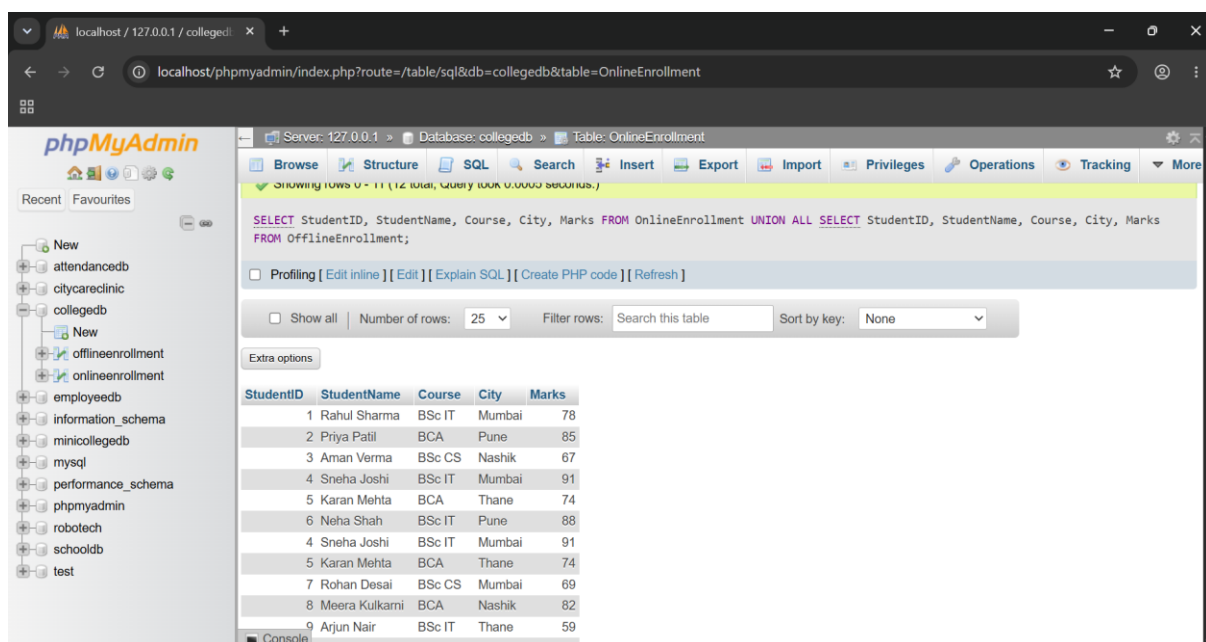
SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

UNION ALL

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;



Q4) Write a MySQL query using INTERSECT to display students who are present in both online and offline enrollment tables.

The output should contain:

Column
StudentID
StudentName

Course
City
Marks

Answer:-

MySQL query

SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

INTERSECT

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;

The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'Table: OnlineEnrollment' is selected. The query editor displays the following SQL query:

```
SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment INTERSECT SELECT StudentID, StudentName, Course, City, Marks FROM OfflineEnrollment;
```

The query results are displayed in a table with 2 rows and 5 columns:

StudentID	StudentName	Course	City	Marks
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74

Q5) Write a MySQL query using EXCEPT to display students who are present only in the online enrollment table.

The output should contain:

Column
StudentID
StudentName

Course
City
Marks

Answer:-

MySQL query

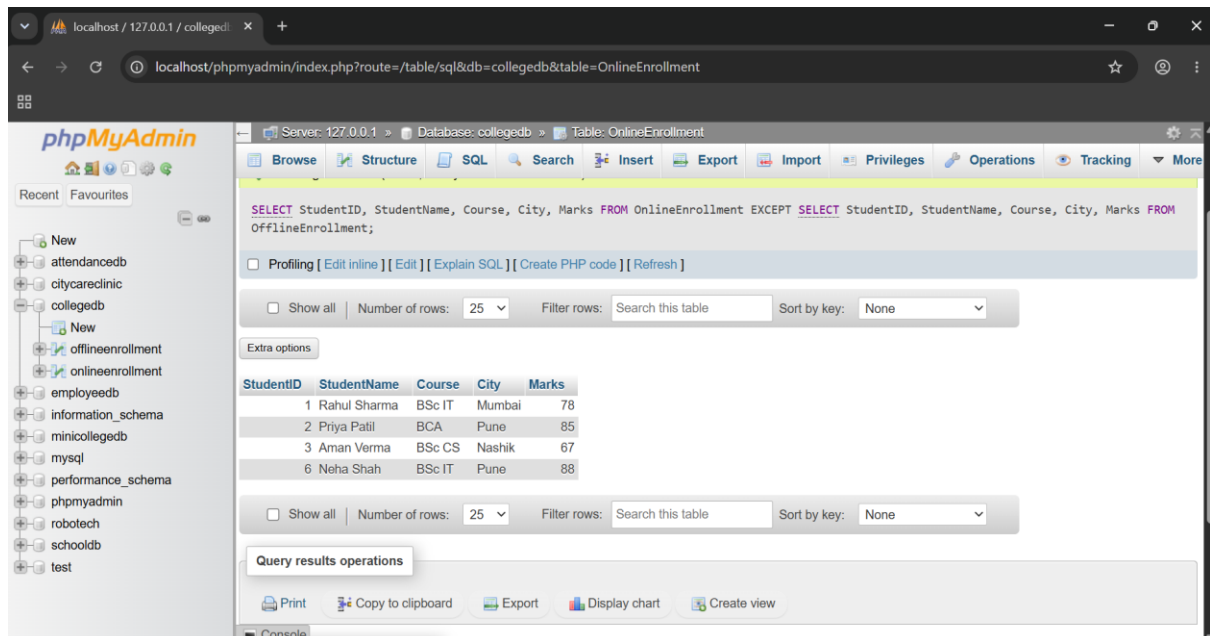
SELECT StudentID, StudentName, Course, City, Marks

FROM OnlineEnrollment

EXCEPT

SELECT StudentID, StudentName, Course, City, Marks

FROM OfflineEnrollment;



Q6) Write a MySQL query to display high-scoring students from both tables.

The query should:

Requirement
Select students with marks greater than or equal to 85
Use UNION to combine both tables
Display StudentID, StudentName, Course, and Marks

Sort the final output by Marks in descending order

Answer:-

SELECT StudentID, StudentName, Course, Marks

FROM OnlineEnrollment

WHERE Marks >= 85

UNION

SELECT StudentID, StudentName, Course, Marks

FROM OfflineEnrollment

WHERE Marks >= 85

ORDER BY Marks DESC;

The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'Table: OnlineEnrollment' is selected. The SQL query entered is: `SELECT StudentID, StudentName, Course, Marks FROM OnlineEnrollment WHERE Marks >= 85 UNION SELECT StudentID, StudentName, Course, Marks FROM OfflineEnrollment WHERE Marks >= 85 ORDER BY Marks DESC;`. The query results are displayed in a table with 4 columns: StudentID, StudentName, Course, and Marks. The results are sorted by Marks in descending order. The table contains 4 rows of data.

StudentID	StudentName	Course	Marks
10	Divya More	BCA	93
4	Sneha Joshi	BSc IT	91
6	Neha Shah	BSc IT	88
2	Priya Patil	BCA	85